

S.A. ENGINEERING COLLEGE

Autonomous Institution | Affiliated to Anna University | NBA & NAAC 'A' Grade

Dept of Electronics & Communication Engg & Dept of Electronics Engg [VLSI Design & Technology]

KERNEL PRIME '26 - SOFTWARE TRACK

National Level 24-Hour Hackathon | October 8-9, 2026

OFFICIAL PROBLEM STATEMENT: PS-03

Adaptive Urban Traffic & Autonomous Emergency Corridor Routing

Domain: Smart Cities / Computer Vision / Graph Optimization

1. PROBLEM OVERVIEW & CHALLENGE STATEMENT

Design a dynamic city-scale routing protocol leveraging real-time vision sensor streams to clear green wave corridors for first responders while minimizing overall grid congestion.

2. MANDATORY ARCHITECTURAL REQUIREMENTS

[REQ-1] YOLO/Edge-vision vehicle counting and density estimation across multi-lane junctions.

[REQ-2] Dynamic reinforcement-learning phase timing for smart traffic light controllers.

[REQ-3] Ultra-reliable ambulance and fire responder priority corridor preemption protocol.

[REQ-4] Decentralized V2X (Vehicle-to-Infrastructure) alert broadcast network.

[REQ-5] Interactive 3D digital-twin GIS dashboard visualizing live congestion and ETA metrics.

3. HACKATHON EVALUATION MATRIX

* System Architecture & Concurrency Resilience: 25%

* Technical Innovation & Production Viability: 25%

* Live Demo & Real-World Latency Benchmarks: 30%

* Security Hardening, UI Polish & Documentation: 20%

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